

NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY
FACULTY OF DATA SCIENCE & ARTIFICIAL INTELLIGENCE (FDA)

OFFICIAL MIDTERM EXAMINATION
Academic Year: 2026–2027
Semester: Fall 2026 Course Code:
DSAI1003

FUNDAMENTAL PROGRAMMING CONCEPTS IN PYTHON

PROGRESS EXAM: BASIC INPUT/OUTPUT IN PYTHON

Paper Code: _____ | Duration: 60 Minutes (Excluding paper distribution)

Candidate Full Name:

Student ID (MSSV): ...

Class / Cohort:

Exam Room / Seat No.:

Section	A	B	C	D	Total (100)	Examiner Signature
Max Score	20	25	25	30	100	
Score						

Instructions: Closed-book examination. All electronic devices and internet access are prohibited. Answer all questions directly in the designated spaces. In programming tasks, use the **Input** → **Process** → **Output** pattern. Do **not** define user functions (def) or use loops/branching. Assume users enter data in the required format. Do not detach pages.

MULTIPLE-CHOICE ANSWER SHEET (Write A, B, C, or D in the corresponding box):

Question	1	2	3	4	5	6	7	8	9	10
Answer										

Section A: Basic Input/Output Concepts

(20 points)

Answer all 10 questions. Each correct answer carries 2 points. Suggested time: 12 minutes.

Q1. Which built-in Python function is primarily used to display data on the screen?

- A. input()
- B. print()
- C. type()
- D. float()

Q2. What is printed by `print(10, 20)` using the default behavior of `print()`?

- A. 1020
- B. 10,20
- C. 10 20
- D. 10:20

Q3. What is the purpose of the `sep` parameter in `print()`?

- A. It changes the data type of printed values.
- B. It controls the separator placed between multiple printed values.
- C. It controls whether the user may enter data.
- D. It rounds floating-point values automatically.

Q4. What is the exact output of `print("A", "B", "C", sep="-")`?

- A. A B C
- B. A-B-C
- C. ABC-
- D. A,B,C

Q5. What is the primary effect of specifying `end=" "` in a `print()` statement?

- A. It places a space after the output instead of the default newline.
- B. It places spaces between all arguments.
- C. It terminates the Python program.
- D. It converts the output into a string.

- Q6.** What data type does `input()` return, even when the user types digits such as 18?
- `int`
 - `float`
 - `str`
 - The type is selected automatically from the typed value.

- Q7.** Suppose the user enters 10 and then 20. What does the following program print?

```
x = input("x = ")
y = input("y = ")
print(x + y)
```

- 30
 - 1020
 - 10 20
 - An error occurs before input is read.
- Q8.** Which statement correctly reads a decimal price from the keyboard for use in arithmetic?
- `price = input(float("Enter price: "))`
 - `price = float(input("Enter price: "))`
 - `price = int("Enter price: ")`
 - `price = print(input("Enter price: "))`
- Q9.** What does `:.2f` mean in an f-string such as `f"Price: {price:.2f}"`?
- Convert the value to an integer with two digits.
 - Display the floating-point value with two digits after the decimal point.
 - Repeat the value twice.
 - Add two to the value before displaying it.
- Q10.** If `x = 10`, what is printed by `print("x")`?
- 10
 - x
 - "x"
 - int

Section B: Output Tracing, Type Conversion & Error Diagnosis

(25 points)

Answer all 5 questions. Each question carries 5 points. Suggested time: 15 minutes.

Q11. Exact Output with `sep` and `end` (5 points)

Predict the **exact screen output**, including spaces and line breaks:

```
print("Python", "I/O", sep="::", end=" | ")
print("Input", "Process", "Output", sep=" -> ")
print("Done")
```

Output Box for Q11

Q12. Strings versus Numbers after input () (5 points)

Assume the user enters 5 for each prompt. Determine the exact output of each part.

```
a = input("a = ")
print(a * 3)

b = int(input("b = "))
print(b * 3)
```

- Output of `print(a * 3)`: _____
- Output of `print(b * 3)`: _____
- Briefly explain why the two outputs differ: _____

Q13. Diagnose and Correct Common Input Errors (5 points)

For each case, state the problem and write a corrected statement where appropriate.

Code / Situation	Problem	Correction / Result
age = input("Age: ") next_age = age + 1		
age = int(input("Age: ")) User enters eighteen		
x = 10 print("x") Goal: display the value of x		

Q14. f-String Tracing and Formatting (5 points)

Predict the exact output:

```
product = "Notebook"
price = 15.5
quantity = 3
total = price * quantity
print(f"Product: {product}")
print(f"Quantity: {quantity}")
print(f"Total: {total:.2f}")
```

Output Box for Q14

Q15. Identify Input, Process, and Output (5 points)

Consider the program:

```
celsius = float(input("Celsius: "))
fahrenheit = 9 / 5 * celsius + 32
print(f"Fahrenheit: {fahrenheit:.2f}")
```

Identify the corresponding components of the Input → Process → Output model:

- **Input:** _____
- **Process:** _____
- **Output:** _____

Section C: Program Design Using the Input → Process → Output Model

(25 points)

Answer all tasks. Suggested time: 15 minutes.

Problem Specification: Student Score Average Report

A program receives three floating-point scores from the user, calculates their arithmetic average, and displays the result with exactly two digits after the decimal point.

Task 1: Identify the IPO Components (6 points)

- **Input(s):** _____
- **Process:** _____
- **Output:** _____

Task 2: Write the Program Logic in Ordered Steps (7 points)

Write clear sequential steps using the Input → Process → Output structure. Your steps should state the required type conversion and output formatting.

Program-Logic Work Area

Task 3: Hand Trace (6 points)

For scores **7.5**, **8.0**, and **9.0**, show the calculation and the exact formatted result.

Hand-Trace Calculation

Task 4: Explain Two Design Choices (6 points)

- a) Why should each score be converted with `float()` rather than used directly from `input()`?
- b) Why is `:.2f` useful when displaying the average score?

Section D: Applied Programming in Python

(30 points)

Write complete, syntactically correct Python statements using the **Input** → **Process** → **Output** pattern. Do **not** define custom functions (`def`), use loops, or use branching. Suggested time: 18 minutes.

Problem 1: Purchase Invoice Calculator (15 points)

Write a Python program that creates a simple purchase summary.

Requirements:

1. Read the `product_name` using `input()`.
2. Read the `unit_price` and convert it to `float`.
3. Read the `quantity` and convert it to `int`.
4. Compute `total = unit_price * quantity`.
5. Display an invoice using f-strings. Show both `unit_price` and `total` with exactly **2 digits after the decimal point**.

The output should follow this general form:

```
Product: Notebook
Unit price: 15.50
Quantity: 3
Total: 46.50
```

Python Solution for Problem 1

Problem 2: Convert Seconds into Minutes and Remaining Seconds (15 points)

Write a Python program that converts a total number of seconds into full minutes and remaining seconds.

Requirements:

1. Prompt the user for `total_seconds` and convert the input to `int`.
2. Compute the number of full minutes using integer floor division (`//`).
3. Compute the remaining seconds using the remainder operator (`%`).
4. Display the original number of seconds, the full minutes, and the remaining seconds clearly.

For example, if the user enters 135, the program should produce values equivalent to:

```
Seconds: 135
Minutes: 2
Remaining seconds: 15
```

Python Solution for Problem 2

— END OF EXAM PAPER —